

Department of Mathematics, IIT Madras
Linear Algebra for Engineers (MA2031)
Assignment-1 (Vector spaces)

1. In each of the following a nonempty set V is given and some operations are defined. Check whether V is a vector space with these operations. Whenever V is a vector space, write explicitly the zero vector and the additive inverse of any vector $v \in V$.

(a) $V = \{(a, b) \in \mathbb{R}^2 : 2a + 3b = 0\}$ with $+$ and $\cdot$ as in $\mathbb{R}^2$. *Ans:* It is.

(b) $V = \{(a, b) \in \mathbb{R}^2 : a + b = 1\}$ with $+$ and $\cdot$ as in $\mathbb{R}^2$. *Ans:* It is not.

(c) $V = \mathbb{R}^2$ with $+$ as in $\mathbb{R}^2$, and $\cdot$ defined by

$$0 \cdot (a, b) = (0, 0), \text{ and for } \alpha \neq 0, \alpha \in \mathbb{R}, \alpha \cdot (a, b) = (a/\alpha, \alpha b).$$

Ans: It is not.

(d) $V = \mathbb{R}_+$, the set of all positive real numbers, with addition $\oplus$ and scalar multiplication $\odot$ defined by $x \oplus y = xy$, and $\alpha \odot x = x^\alpha$ for $x, y \in V$ and $\alpha \in \mathbb{R}$. *Ans:* It is.

(e) $V = \mathbb{Q}$, the set of all rational numbers, with $+$ and $\cdot$ as in $\mathbb{R}$.

Ans: It is not.

(f) $V = \{(a, b) \in \mathbb{R}^2 : a + b = 1\}$ with addition $\oplus$ and scalar multiplication $\odot$ as $(a, b) \oplus (c, d) = (a + c - 1, b + d)$, and $\alpha \odot (a, b) = (\alpha a - \alpha + 1, \alpha b)$.

Ans: It is.

2. Let V be a vector space over $\mathbb{F}$. Show the following:

(a) For all $x, y, z \in V$, $x + y = z + y$ implies $x = z$.

(b) Let $\alpha, \beta \in \mathbb{F}, x \in V, x \neq 0$. Then, $\alpha x \neq \beta x$ iff $\alpha \neq \beta$.

(c) If V has two vectors, then V has an infinite number of vectors.

3. In each of the following, a vector space V and a subset U of V are given. Check whether U is a subspace of V .

(a) $V = \mathbb{R}^2, U = \{(a, b) \in V : b = 2a - \alpha\}$ for some $\alpha \neq 0$. *Ans:* It is not

(b) $V = \mathbb{R}^3, U = \{(a, b, c) \in V : 2a - b - c = 0\}$. *Ans:* It is.

(c) $V = \mathbb{F}_3[t], U = \{a + bt + ct^2 + dt^3 \in V : a = 0\}$. *Ans:* It is.

(d) $V = C[0, 1], U = \{f \in V : f(t) \geq 0 \text{ for all } t \in [0, 1]\}$. *Ans:* It is not.

4. Let S be a nonempty set and let $s \in S$. Let V be the set of all functions $f : S \rightarrow \mathbb{R}$ with $f(s) = 0$. Is V a vector space over $\mathbb{R}$ with the usual addition and scalar multiplication of functions?

Ans: It is a subspace of the space of all functions from S to $\mathbb{R}$.

5. Do the polynomials $t^3 - 2t^2 + 1$, $4t^2 - t + 3$, and $3t - 2$ span $\mathbb{F}_3[t]$?

Ans: No.

6. Let $u_1(t) = 1$, and for $n = 2, 3, \dots$, let $u_n(t) = 1 + t + \dots + t^{n-1}$. Show that $\{u_1, \dots, u_n\}$ spans $\mathbb{F}_{n-1}[t]$. Is it true that $\{u_1, u_2, \dots\}$ spans $\mathbb{F}[t]$? *Ans:* Yes.

7. We know that $e^t = 1 + t + \frac{1}{2!}t^2 + \dots$ for each $t \in \mathbb{R}$. Does it imply that $e^t \in \text{span}\{1, t, t^2, \dots\}$?

Ans: No.

8. Let A and B be subsets of a vector space V . Prove (a)-(c) and give counter examples for (d)-(f):

(a) $\text{span}(\text{span}(A)) = \text{span}(A)$.

(b) If $A \subseteq B$, then $\text{span}(A) \subseteq \text{span}(B)$.

(c) $\text{span}(A \cap B) \subseteq \text{span}(A) \cap \text{span}(B)$.

(d) $\text{span}(A) \cap \text{span}(B) \subseteq \text{span}(A \cap B)$.

(e) $\text{span}(A) \setminus \text{span}(B) \subseteq \text{span}(A \setminus B)$.

(f) $\text{span}(A \setminus B) \subseteq (\text{span}(A) \setminus \text{span}(B)) \cup \{0\}$.

9. Give suitable real vector spaces U, V, W so that $U + V = U + W$ but $V \neq W$.

10. In each of the following, a vector space V and a subset S of V are given. Determine whether S is linearly dependent; and if it is, express one of the vectors in S as a linear combination of the remaining vectors.

(a) $V = \mathbb{R}^3$, $S = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$. Ans: LD.

(b) $V = \mathbb{C}^3$, $S = \{(1, -3, -2), (-3, 1, 3), (2, 5, 7)\}$. Ans: LI.

(c) $V = \mathbb{F}_3[t]$, $S = \{t^2 - 3t + 5, t^3 + 2t^2 - t + 1, t^3 + 3t^2 - 1\}$. Ans: LI.

(d) $V =$ the set of all functions from $\mathbb{R}$ to $\mathbb{R}$, $S = \{2, \sin^2 t, \cos^2 t\}$.

Ans: LD.

(e) $V =$ the set of all functions from $\mathbb{R}$ to $\mathbb{R}$, $S = \{1, \sin t, \sin 2t\}$.

Ans: LI.

11. Prove the following:

(a) Each superset of a linearly dependent set is linearly dependent.

(b) Each subset of a linearly independent set is linearly independent.

(c) Union of any two linearly dependent sets is linearly dependent.

(d) Intersection of any two linearly independent sets is linearly independent.

12. Construct examples to show that the following statements are false:

(a) Each subset of a linearly dependent set is linearly dependent.

(b) Each superset of a linearly independent set is linearly independent.

(c) Union of any two linearly independent sets is linearly independent.

(d) Intersection of any two linearly dependent sets is linearly dependent.

13. Let A and B be subsets of a vector space. Prove or disprove:

(a) If $\text{span}(A) \cap \text{span}(B) = \{0\}$, then $A \cup B$ is linearly independent.

(b) If $A \cup B$ is linearly independent, then $\text{span}(A) \cap \text{span}(B) = \{0\}$.

Ans: (a) disprove, (b) prove.

14. Let $u, v, w, x, y_1, y_2, y_3, y_4, y_5$ be vectors in a vector space V satisfying the relations:

$$y_1 = u + v + w, y_2 = 2v + w + x, y_3 = u + 3w + x, y_4 = 2u + v + 4x, \text{ and } y_5 = u + 2v + 3w + 4x.$$

Are the vectors $y_1, \dots, y_5$ linearly dependent or independent? Ans: LD.

15. Determine which of the following sets are bases of $\mathbb{F}_2[t]$?

(a) $\{1, 1+t, 1+t+t^2\}$. (b) $\{1+2t+3t^2, 2-5t+3t^2, 3t+t^2\}$.

Ans: (a) Basis (b) Not a basis.

16. Let $\{x, y, z\}$ be a basis of a vector space V . Are the following bases of V ?

(a) $\{x+y, y+z, z+x\}$ (b) $\{x-y, y-z, z-x\}$

Ans: (a) Yes. (b) No.

17. Find a basis for the subspace $\{(a, b, c) \in \mathbb{R}^3 : a + b + c = 0\}$ of $\mathbb{R}^3$.

18. Extend the set $\{1+t^2, 1-t^2\}$ to a basis of $\mathbb{F}_3[t]$.

19. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:

(a) $\{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_1 - a_3 - a_4 = 0\}$. Ans: 4.

(b) $\{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_2 = a_3 = a_4, a_1 + a_5 = 0\}$. Ans: 2.

20. Show that if U and W are subspaces of $\mathbb{R}^9$ with $\dim(U) = 5 = \dim(W)$, then $U \cap W \neq \{0\}$.

21. Let $U = \text{span}\{(1, 2, 3), (2, 1, 1)\}$ and $W = \text{span}\{(1, 0, 1), (3, 0, -1)\}$. Find a basis for $U \cap W$. Also, find $\dim(U + W)$.

Ans: $\dim(U \cap W) = 1$, $\dim(U + W) = 3$.

22. In $\mathbb{R}^3$, what is $\dim(\text{span}\{e_1 + e_2, e_2 + e_3, e_3 + e_1\})$? Ans: 3.

23. Find a basis and dimension of the subspace of $\mathbb{R}^5$ that is spanned by the vectors $(1, -1, 0, 2, 1)$, $(2, 1, -2, 0, 0)$, $(0, -3, 2, 4, 2)$, $(3, 3, -4, -2, -1)$, $(2, 4, 1, 0, 1)$, $(5, 7, -3, -2, 0)$. Ans: $\dim = 4$.

24. Determine a basis for $\text{span}\{1 + t^2, -1 + t + t^2, -6 + 3t, 1 + t^2 + t^3, t^3\}$.

Ans: $\dim = 3$.

25. Let $U = \{(a, b, c, d) \in \mathbb{R}^4 : b = -a\}$ and $W = \{(a, b, c, d) \in \mathbb{R}^4 : c = -a\}$. Find the dimensions of the subspaces U , W , $U + W$ and $U \cap W$ of $\mathbb{R}^4$.

Ans: $\dim(U) = 3$, $\dim(W) = 3$, $\dim(U \cap W) = 2$, $\dim(U + W) = 4$.